

# A Paired-Zero Subcase of the Brändén–Chasse Fourier Transform Conjecture

Candidate partial result for arXiv:1402.2795

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**Status.** Candidate partial result, likely valid. The unrestricted conjecture is not claimed.

## Abstract

Brändén and Chasse conjecture that if a real transcendental entire function  $G$  has derivative in the Laguerre–Pólya class and decays appropriately on the imaginary axis, then the Fourier transform of  $e^{G(it)}$  is again in the Laguerre–Pólya class. We prove the conjecture when the zeros of  $G'$  admit a summable positive/negative pairing whose total imaginary-axis angular defect is less than  $\pi/2$ . The proof constructs real-rooted polynomial derivative approximants and obtains a uniform Gaussian-or-better tail estimate, which is the compactness input missing in the unrestricted problem.

## 1 Source question

The target is Conjecture 6.1 on page 29 of Brändén–Chasse [1]. The source statement is reproduced below.

- (1) Obtain a characterization of real strip preservers which either uses properties of the operator symbol, or some other readily verifiable criteria.
- (2) Determine if operators arising from Theorem 1.8 which possess property (I) of a strong universal factors also satisfy property (II).
- (3) Given  $f \in \mathcal{L}\text{-}\mathcal{P}$  of order one and mean type, determine a partner  $g \in \mathcal{L}\text{-}\mathcal{P}$  such that  $(f + ig)(D)$  has the form of  $T$  in Theorem 1.8 (note the zeros of  $f$  and  $g$  must interlace). Do all operators with order strictly less than two, and property (I) of strong universal factors, have this form?
- (4) Prove the following conjecture,

**Conjecture 6.1.** *Suppose  $G(t) = \sum_{k=0}^{\infty} \gamma_k x^k / k!$  is a real transcendental entire function,  $G' \in \mathcal{L}\text{-}\mathcal{P}$ , and  $G(it) \rightarrow -\infty$  as  $t \rightarrow \pm\infty$ . Then*

$$F(z) = \int_{-\infty}^{\infty} e^{G(it)} e^{izt} dt \in \mathcal{L}\text{-}\mathcal{P}.$$

## REFERENCES

- [1] R. Beals, R. Wong, *Special Functions*, Cambridge University Press, NY, 2010.
- [2] D. Bleecker, G. Csordas, *Hermite expansions and the distribution of zeros of entire functions*, Acta Sci. Math. (Szeged) **67** (2001), 177–196.
- [3] J. Borcea, P. Brändén, *Pólya–Schur master theorems for circular domains and their boundaries*, Ann. of Math. **170** (2009), 465–492.
- [4] J. Borcea, P. Brändén, *The Lee Yang and Pólya–Schur programs. I. Linear operators on*

The displayed source uses the notation  $G(it) \rightarrow -\infty$ . Since complex numbers are not ordered, and since Corollary 5.5 and Proposition 5.6 directly before the conjecture use  $\operatorname{Re} G(it) \rightarrow -\infty$ , we

treat the nontrivial intended reading as

$$\operatorname{Re} G(it) \longrightarrow -\infty \quad (t \rightarrow \pm\infty).$$

Recall that  $\mathcal{LP}$  is the class of real entire functions that are locally uniform limits of real polynomials having only real zeros.

## 2 Candidate partial theorem

**Theorem 1** (paired genus-zero subcase). *Let  $m \in \mathbb{Z}_{\geq 0}$ ,  $\alpha \geq 0$ , and  $c \in \mathbb{R}$  satisfy  $c(-1)^m > 0$ . Let  $a_j, b_j > 0$  satisfy*

$$\sum_{j=1}^{\infty} \left( \frac{1}{a_j} + \frac{1}{b_j} \right) < \infty \quad (1)$$

and

$$A := \sum_{j=1}^{\infty} \arctan \left( \frac{|a_j - b_j|}{2\sqrt{a_j b_j}} \right) < \frac{\pi}{2}. \quad (2)$$

Define

$$H(z) = cz^{2m+1} e^{-\alpha z^2} \prod_{j=1}^{\infty} \left( 1 - \frac{z}{a_j} \right) \left( 1 + \frac{z}{b_j} \right), \quad (3)$$

and let  $G$  be the real entire primitive of  $H$  with a prescribed real value  $G(0)$ . Then

$$\operatorname{Re} G(it) \leq G(0) - \frac{|c| \cos A}{2m+2} |t|^{2m+2} \quad (t \in \mathbb{R}). \quad (4)$$

Consequently

$$F(z) = \int_{-\infty}^{\infty} e^{G(it)} e^{izt} dt$$

is an entire function in  $\mathcal{LP}$ .

Condition (1) makes (3) a convergent genus-zero canonical product. In particular  $H \in \mathcal{LP}$ . The conclusion therefore proves the source conjecture for an infinite-dimensional, explicitly checkable paired-zero class.

## 3 Idea of the proof

For one paired positive/negative zero, substitution  $z = it$  produces a factor with positive real part. Its argument can deviate from zero by at most  $\arctan(|a_j - b_j|/(2\sqrt{a_j b_j}))$ . The strict total bound (2) therefore keeps every finite canonical-product truncation of  $H(it)$  inside the same open half-plane. Its imaginary part has the sign of  $t$  and a uniform polynomial lower bound.

After integration, this becomes one coercive upper bound for the real parts of all polynomial primitives. Brändén–Chasse’s polynomial theorem applies to each primitive. The common coercive bound is strong enough for dominated convergence of the Fourier transforms, so the Laguerre–Pólya conclusion survives the transcendental limit.

## 4 Proof

For  $N \geq 1$ , put

$$H_N(z) = cz^{2m+1} \left(1 - \frac{\alpha z^2}{N}\right)^N \prod_{j=1}^N \left(1 - \frac{z}{a_j}\right) \left(1 + \frac{z}{b_j}\right). \quad (5)$$

(When  $\alpha = 0$ , the middle factor is 1.) Every  $H_N$  is a real polynomial with only real zeros, and  $H_N \rightarrow H$  locally uniformly. Let  $G_N$  be its primitive normalized by  $G_N(0) = G(0)$ . Then  $G_N \rightarrow G$  locally uniformly.

Fix  $t > 0$ . The  $j$ th paired factor in (5) equals

$$\left(1 - \frac{it}{a_j}\right) \left(1 + \frac{it}{b_j}\right) = 1 + \frac{t^2}{a_j b_j} + it \left(\frac{1}{b_j} - \frac{1}{a_j}\right).$$

Its real part is positive. If  $\theta_j(t)$  is its principal argument, then

$$|\tan \theta_j(t)| = \frac{t|a_j - b_j|}{a_j b_j + t^2} \leq \frac{|a_j - b_j|}{2\sqrt{a_j b_j}},$$

because  $t/(a_j b_j + t^2)$  is maximized at  $t = \sqrt{a_j b_j}$ . Hence

$$|\theta_j(t)| \leq \arctan \left( \frac{|a_j - b_j|}{2\sqrt{a_j b_j}} \right). \quad (6)$$

The factor  $(1 + \alpha t^2/N)^N$  is positive. Moreover  $c(it)^{2m+1} = i|c|t^{2m+1}$  because  $c(-1)^m > 0$ . Equations (2) and (6) therefore give

$$\operatorname{Im} H_N(it) \geq |c| \cos(A) t^{2m+1}, \quad t > 0. \quad (7)$$

Indeed, the argument of  $H_N(it)$  differs from  $\pi/2$  by at most  $A$ , while every remaining factor in its modulus is at least 1. Since  $H_N$  has real coefficients, conjugation gives the corresponding negative sign for  $t < 0$ .

Along the imaginary axis,

$$\frac{d}{dt} \operatorname{Re} G_N(it) = \operatorname{Re}(i H_N(it)) = -\operatorname{Im} H_N(it).$$

Integrating (7) on either side of the origin yields the uniform estimate

$$\operatorname{Re} G_N(it) \leq G(0) - q|t|^{2m+2}, \quad q := \frac{|c| \cos A}{2m+2} > 0. \quad (8)$$

Passing to the pointwise limit gives (4) for  $G$ .

Each  $G_N$  is a real polynomial,  $G'_N = H_N \in \mathcal{LP}$ , and  $\operatorname{Re} G_N(it) \rightarrow -\infty$  at both ends. Corollary 5.5 of [1] therefore implies

$$F_N(z) := \int_{-\infty}^{\infty} e^{G_N(it)} e^{izt} dt \in \mathcal{LP}.$$

If  $K \subset \mathbb{C}$  is compact and  $M = \sup_{z \in K} |\operatorname{Im} z|$ , then (8) gives the  $N$ -independent integrable majorant

$$|e^{G_N(it)} e^{izt}| \leq \exp(G(0) - q|t|^{2m+2} + M|t|), \quad z \in K.$$

Dominated convergence now yields  $F_N \rightarrow F$  locally uniformly on  $\mathbb{C}$ . The class  $\mathcal{LP}$  is closed under locally uniform limits. The limit is not identically zero, since  $e^{G(it)}$  is a nonzero integrable function and the Fourier transform is injective on  $L^1(\mathbb{R})$ . Thus  $F \in \mathcal{LP}$ , completing the proof.  $\square$

## 5 Verification notes and limitations

The proof uses no numerical claim. The phase estimate, the sign for negative  $t$ , and the primitive derivative were recalculated independently; details are recorded in the companion verification report.

The added hypothesis (2) is the essential limitation. A general Laguerre–Pólya function has locally convergent real-rooted polynomial approximants, but their primitives need not share an integrable imaginary-axis majorant. The source condition controls the accumulated primitive, not the pointwise phase of arbitrary truncations. Thus this packet does not resolve the unrestricted conjecture.

A bounded search of the run indexes, exact web/arXiv phrases, and the indexed citing works of the published source found no later proof of Conjecture 6.1 and no occurrence of criterion (2). This is not a definitive novelty claim. Some functions covered here may also satisfy Proposition 5.6 of the source; strict separation from that coefficient-sign class has not been established.

**Human-review recommendation.** Check first the intended  $\operatorname{Re} G(it)$  interpretation and the use of source Corollary 5.5. Conditional on that textual interpretation, the partial theorem itself should be reviewed as a short canonical-product and dominated- convergence argument.

## References

- [1] P. Brändén and M. Chasse, *Classification theorems for operators preserving zeros in a strip*, J. Anal. Math. **132** (2017), 177–215; arXiv:1402.2795.